

QUIZ 03 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday. Rather, they are designed to give you a sense of the style of multiple choice questions I typically ask. As a reminder (which was sent via Canvas Announcement): the quiz will be 15 questions, 10 of which will be based off this week's material (six step method, chance models, etc.) and 5 of which will be based off last week's material (sampling, interpretation of probabilities, etc.)

Use the following setup for Questions 1 through 4:

Marta claims to be able to tell whether someone has blue eyes or not, just by listening to their voice. When presented with the voices of 12 people, she made a correct identification of blue-eyed status (i.e. whether or not a particular person has blue eyes) 10 times.

1. Which of the following best describes an observational unit in the context of this dataset?

- (A) One person
- (B) One trial
- (C) One trial of Marta guessing whether someone has blue eyes or not**
- (D) The set of people in the world.

Solution: Our data represents observations from a random process, meaning one observational unit is one trial. But, as mentioned in Lecture on Wednesday (September 9), we typically want to phrase an observational unit in the context of the study. As such, even though answer choice (B) is technically correct, **choice (C) is the most correct option** because it not only identifies an observational unit as one trial but also describes what one trial is in the context of the study.

2. What is the variable, and is the variable quantitative or categorical?

- (A) Variable = The number of people in the world who have blue eyes; categorical
- (B) Variable = The number of people in the world who have blue eyes; quantitative
- (C) Variable = Whether Marta identifies the correct blue-eyed status; categorical**
- (D) Variable = Whether Marta identifies the correct blue-eyed status; quantitative

Solution: A variable is just an attribute/characteristic recorded from/on each observational unit. Each observational unit, in the context of this study, is one trial; from each trial we only record “whether or not Marta identified the correct blue-eyed status” which means this is the sole variable in our study. This is a **categorical** variable, with levels “yes” or “no.”

3. What is the sample size?

- (A) 0 (B) 0.8333 (C) 10 **(D) 12**

Solution: The sample size is the number of observational units, which is **12** in this case.

4. What is the observed value of the sample proportion?

- (A) 0 **(B) 0.8333** (C) 10 (D) 12

Solution: The observed value of the sample proportion is the proportion of times, in our sample of size 12, that Marta correctly identified the blue-eyed status; namely **$10/12 \approx 0.8333$** .

5. Which of the following best describes the parameter of interest?

- (A) The long-run proportion of times Marta correctly identifies a person's blue-eyed status**
(B) The proportion of times among 12 trials that Marta correctly identifies a person's blue-eyed status
(C) The proportion of all people in the world who have blue eyes
(D) The proportion of all people in San Luis Obispo whose blue-eyed status Marta would correctly identify.

Solution: When our data takes the form of observations from a random process with binary outcome, the parameter of interest is the long-run proportion of “successes.” In this case, a “success” is Marta correctly identifying the number of boba pearls in a cup of boba. Note that choice (D) is somewhat tempting, but it is not the *best* descriptor as it is missing the *long-run* or *large-sample* interpretation of a long-run proportion.

6. If Marta was just guessing, by random chance, what would be the long-run proportion of times she'd correctly identify the blue-eyed status of a person?

- (A) Less than 0.5
(B) Equal to 0.5
(C) Equal to 0.83333
(D) Greater than 0.5

Solution: If she was just guessing, we'd expect her to, in the long run, correctly identify a person's blue-eyed status 50% of the time.

7. If Marta was just guessing, by random chance, what would be the long-run proportion of times she'd correctly identify the blue-eyed status of a person?
- (A) Less than 0.5
 - (B) Equal to 0.5
 - (C) Equal to 0.83333
 - (D) Greater than 0.5

Solution: If she was just guessing, we'd expect her to, in the long run, correctly identify a person's blue-eyed status 50% of the time.

In addition to the original setup, use the following for questions 8 through 11:

We'll now turn our attention to setting up a simulation of the chance model, which essentially amounts to filling in the following table:

One Coin Flip	
Coin Landing Heads	
Coin landing Tails	
Chance of Heads	
One Repetition	

8. What does one coin flip represent?
- (A) Marta correctly identifying the blue-eyed status of a person
 - (B) Marta incorrectly identifying the blue-eyed status of a person
 - (C) **One trial of Marta guessing the blue-eyed status of a person**
 - (D) A person with blue eyes
 - (E) A person with not-blue eyes

Solution: One coin flip represents one trial of the experiment.

9. What does one coin flip represent?
- (A) Marta correctly identifying the blue-eyed status of a person
 - (B) Marta incorrectly identifying the blue-eyed status of a person
 - (C) **One trial of Marta guessing the blue-eyed status of a person**
 - (D) A person with blue eyes
 - (E) A person with not-blue eyes

Solution: One coin flip represents one trial of the experiment.

10. What does the coin landing heads represent?

- (A) **Marta correctly identifying the blue-eyed status of a person**
- (B) Marta incorrectly identifying the blue-eyed status of a person
- (C) One trial of Marta guessing the blue-eyed status of a person
- (D) A person with blue eyes
- (E) A person with not-blue eyes

Solution: The coin landing “heads” corresponds to a “success.”

11. What should we set the chance of heads to be?

- (A) 0.0
- (B) **0.5**
- (C) 0.8333
- (D) 1.0
- (E) There is Not Enough Information to Determine

Solution: We set the chance of heads to be **0.5**, corresponding to the “just guessing” model.

In addition to the original setup, use the following for questions 12 through 15:

Shown below is a plot from the One Proportion applet representing 250 repetitions of the chance model, with corresponding counts on top of each.

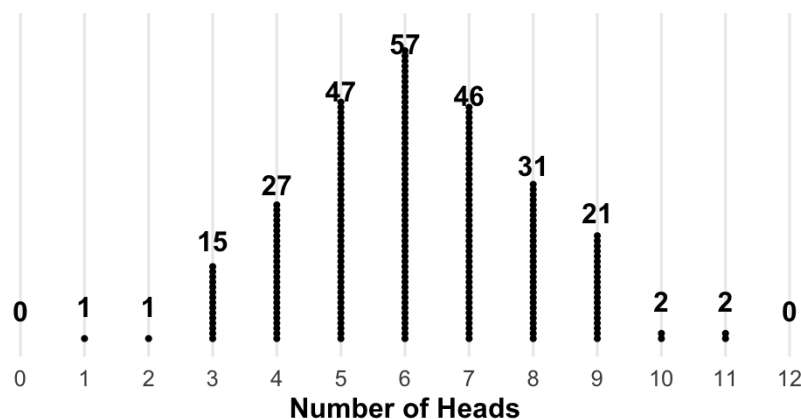


Figure 1: A dotplot representing 250 repetitions of the chance model

12. What does each dot on the dotplot represent?

- (A) One trial
- (B) One sample of 10 trials
- (C) One sample of 12 trials**
- (D) One sample of 250 trials

Solution: This question combines two concepts: first is recognizing that each dot represents a sample, and the location of each dot corresponds to the number of heads. The second is recognizing what does in the last row of the table above (“One repetition”) - recall that we always set the number of trials per repetition to equal the sample size (which, in this case, is 12).

13. Based on this figure, what would you say is the more plausible conclusion?

- (A) Marta is just guessing, and she correctly identified 10 out of 12 people’s blue-eyed statuses by pure chance alone.
- (B) Marta is doing something more than just guessing, indicating that the long-run proportion of times she correctly identifies a person’s blue-eyed status just by listening to their voice is greater than 0.5.**

Solution: 11 appears quite infrequently on the dotplot; hence, it is pretty unlikely to observe a sample count of 11 if she were just guessing. As such, it seems more plausible that she’s doing something more than just guessing.
